

Cloud Systems Administration

Reference Guide — Personalized for Your Current Cert Stack

What is a Cloud Systems Administrator? You manage, monitor, secure, and automate cloud infrastructure — virtual machines, storage, networking, identity, and the tools that connect them. In 2026 this role spans AWS, Azure, and/or Google Cloud, and increasingly requires knowing Linux, scripting (Python/Bash), infrastructure-as-code (Terraform), containers (Docker/Kubernetes), and monitoring tools. Median pay: \$140,000/year (Glassdoor, Feb 2026).

Your Current Certs — You're already ahead of most beginners: CompTIA A+ ✓ | CompTIA Security+ ✓ | CompTIA Cloud+ ✓ | Linux Essentials ✓ | AWS Cloud Practitioner ✓ | Azure Fundamentals (AZ-900) ✓

Where You Stand: You've cleared all the foundational certs. The CompTIA trifecta (A+/Sec+/Cloud+) plus both vendor fundamentals puts you solidly past entry level. Your next moves are the hands-on admin certs (SOA-C02 / AZ-104), then Infrastructure as Code and containers. That's the stack that gets you to mid-level cloud admin and \$100K+ roles.

1 — Certifications: What's Next For You

The certs below are where you go from here — skipping everything you already hold. They're ordered by priority for a Cloud SysAdmin career path. Certs marked **PRIORITY** are the ones hiring managers in NoVA/DC look for first.

Note on Cost: Most vendor exams run \$150–\$300. Many NoVA employers (especially government contractors) reimburse exam fees. Always ask before paying out of pocket. Also check if GMU has any voucher programs through their IT department.

★ **PRIORITY — AWS SysOps Administrator Associate (SOA-C02)**

The core AWS hands-on admin cert. Covers deploying, managing, monitoring, and securing AWS workloads in production. This is the direct next step from AWS Cloud Practitioner and the cert most AWS cloud admin job postings require. ~\$300 exam.

★ **PRIORITY — Microsoft Azure Administrator (AZ-104)**

The core Azure hands-on admin cert. Covers VMs, storage, networking, identity (Active Directory), and monitoring. Direct step up from AZ-900. Most Azure sysadmin roles require this. ~\$165 exam.

CompTIA Network+ (if not already held)

If you don't have Network+ yet, add it — TCP/IP, subnetting, DNS, VPNs, and firewalls underpin everything in cloud networking. Pairs well with Cloud+. Check if your A+ study covered this already.

HashiCorp Terraform Associate

Infrastructure-as-code cert. Validates you can provision and manage cloud resources using Terraform. Increasingly listed as required in mid-level cloud admin job postings. ~\$70 exam.

Linux Professional Institute LPIC-1

Upgrade from Linux Essentials to LPIC-1. Goes deeper: file systems, process management, networking, shell scripting, and package management. Most cloud VMs run Linux — this validates real admin-level Linux skills.

AWS Solutions Architect Associate (SAA-C03)

Pairs well with SOA-C02. Covers designing scalable, highly available AWS architectures. Opens architect-track roles in addition to admin roles. Often studied alongside or right after SysOps. ~\$300 exam.

Microsoft Azure Administrator Expert (AZ-305)

Senior-level Azure cert. Requires AZ-104 as a prerequisite. Covers designing Azure solutions at enterprise scale. Good 12–18 month target after AZ-104.

Certified Kubernetes Administrator (CKA)

Hands-on K8s cert — it's a practical exam, not multiple choice. You administer a live Kubernetes cluster under time pressure. High value in DevOps-adjacent cloud roles. ~\$395 exam.

AWS Advanced Networking Specialty (ANS-C01)

Deep networking cert for AWS — VPCs, Direct Connect, Transit Gateway, hybrid architectures. High value in NoVA given the density of enterprise/government AWS deployments. Best after SOA-C02.

CompTIA CASP+ (CAS-004)

Advanced Security Practitioner — the advanced-level DoD 8570 cert above Security+. Opens IAT Level III and IAM roles with government contractors. High value in the DC/NoVA corridor if you want to stay in the security lane.

2 — Core Skills & Tools to Learn

These are the hands-on technologies you need to actually do the job. Having cloud fundamentals certs is the door opener — knowing these tools is what keeps you employed and growing. Each resource below is the best starting point for that skill.

Terraform — Official Tutorials (Free)

HashiCorp's official free tutorials. Start with 'Get Started — AWS' or Azure. Write your first config that spins up real infrastructure. This is the most important new skill to add to your stack right now.

AWS Skill Builder — SysOps Path (Free)

1,000+ free AWS courses. Filter for 'SysOps Administrator' learning path. Directly maps to SOA-C02 exam content. Use this as your primary SOA-C02 study resource.

Microsoft Learn — AZ-104 Path (Free)

Free official AZ-104 study path from Microsoft. Includes sandbox labs — no Azure account or credit card needed. Your primary free resource for AZ-104 prep.

Docker — Getting Started (Free)

Official Docker tutorials. Containers are everywhere in cloud environments. Start here — understanding Docker is the prerequisite for Kubernetes.

Kubernetes — Official Tutorials (Free)

Official K8s interactive tutorials. Container orchestration at scale. Follow the basics tutorial first, then work toward CKA prep with Killercode labs.

Ansible — Getting Started (Free)

Configuration management and automation tool. Automates server setup, patching, and config at scale. Terraform builds the infrastructure, Ansible configures it.

Prometheus + Grafana (Free)

The industry-standard monitoring stack. Prometheus scrapes metrics, Grafana visualizes them. You will configure this in virtually every cloud environment you manage.

Git — Official Docs (Free)

Version control for all your IaC configs, scripts, and runbooks. Non-negotiable. If your Terraform code isn't in Git, it doesn't exist.

Bash Scripting Guide (Free)

Free guide to Bash shell scripting. Automate repetitive admin tasks, write deployment scripts, and build tooling. Pairs with your Linux Essentials background.

KodeKloud — Hands-On Labs (Paid ~\$15/mo)

Browser-based labs for Kubernetes, Docker, Terraform, Ansible, and Linux. The best paid lab environment for the price. Worth it when preparing for CKA or Terraform Associate.

3 — Learning Platforms & Communities

Where to study, practice, and ask questions. Free options are flagged. The Reddit communities are underrated — real sysadmins sharing real problems and solutions.

GMU Tip: Check library.gmu.edu — GMU students get free access to LinkedIn Learning and Pluralsight (which includes A Cloud Guru content). Use that before paying for anything.

TutorialsDojo — Practice Exams (Free + Paid)

Best SOA-C02 and AZ-104 practice exams available. Free cheat sheets for every AWS/Azure service. Paid practice tests (~\$15) closely mirror real exam questions. Use this for final exam prep.

A Cloud Guru / Pluralsight (Paid — check GMU)

A Cloud Guru is now fully integrated into Pluralsight under the 'Cloud+' plan. Best cloud cert prep platform — SOA-C02 and AZ-104 courses with hands-on sandbox labs. Check GMU library first (library.gmu.edu) — Pluralsight access may be free through the university.

Killercoda — Free Browser Labs

FREE browser-based labs for Linux, Kubernetes, Docker, Terraform, and CI/CD. No install needed. Best free hands-on lab platform — use this heavily.

Play with Docker (Free)

Free Docker sandbox in your browser. Real Linux environment, Docker pre-installed. 4-hour sessions. Zero setup.

Play with Kubernetes (Free)

Free Kubernetes playground. Real K8s cluster in your browser. Good for following Kubernetes tutorials without managing your own cluster.

AWS Free Tier — 12 Months

Real AWS console, free for 12 months. Practice EC2, VPCs, IAM, S3, CloudWatch — the actual services on SOA-C02. Set a \$5 billing alert so you don't get surprised.

Azure Free Account — 12 Months

Real Azure environment, free for 12 months. Practice everything on AZ-104 in the actual Azure portal. Same billing alert advice applies.

r/sysadmin (Reddit)

500K+ sysadmin community. Real-world Q&A, cert advice, job market intel, and war stories. Read the wiki. Invaluable for staying current on what's actually happening in the field.

TechWorld with Nana — YouTube (Free)

Free YouTube channel. Best video content for Docker, Kubernetes, Terraform, and CI/CD. Watch her playlists in order — Docker first, then Kubernetes, then Terraform.

GitHub — Awesome Sysadmin

Community-maintained list of sysadmin tools, software, and resources. Huge catalog. Useful for discovering tools you didn't know existed.

Linux Foundation — Free Courses

Free Linux and Kubernetes courses from the org that maintains Linux. Vendor-neutral. Includes CKA prep material and cloud infrastructure courses.

freeCodeCamp — YouTube (Free)

Full-length free courses on YouTube: Docker, Kubernetes, Terraform, Linux, and AWS. Some are 10+ hours. Completely free.

Your Learning Path — Starting From Where You Are

You've already completed Steps 1–3 that most people start with. Your path begins at Step 4.

✓ **DONE — Foundation:** CompTIA A+, Security+, Cloud+, Linux Essentials, AWS Cloud Practitioner, Azure AZ-900. Solid foundation. You're ahead of most people entering this field.

Step 1 — Hands-On AWS Admin (SOA-C02): Your most impactful next cert. Study FREE with AWS Skill Builder SysOps path + TutorialsDojo practice exams. Spin up real labs on AWS Free Tier. This cert + your existing stack is hireable.

Step 2 — Hands-On Azure Admin (AZ-104): Study FREE with Microsoft Learn AZ-104 path (includes sandbox labs). Use Azure Free Account for practice. TutorialsDojo for practice exams. Earning both SOA-C02 and AZ-104 makes you multi-cloud hireable.

Step 3 — Infrastructure as Code (Terraform): Learn Terraform FREE via HashiCorp's official tutorials. Write configs that deploy real AWS/Azure infrastructure. Take the Terraform Associate cert. This is the skill gap most cert-holders have.

Step 4 — Containers: Docker first (Play with Docker, free), then Kubernetes (KillerCoda, free). Watch TechWorld with Nana's playlists on YouTube. This is the stack separating junior from mid-level cloud admins.

Step 5 — Scripting: Build on your Linux Essentials background with Bash scripting and Python. Automate your admin tasks. Write scripts that interact with AWS/Azure APIs. This is what makes you a force multiplier.

Step 6 — Specialize (12–18 month target): Choose your lane: DevOps (add CKA + CI/CD), Security (add CASP+ or cloud security cert), or Architecture (add AWS Solutions Architect Associate). The NoVA market rewards specialization.

Tech Stack to Add: Terraform (IaC) · Docker · Kubernetes · Ansible · Bash/Python scripting · Git · Prometheus/Grafana · CI/CD pipelines

You already have: Linux · AWS fundamentals · Azure fundamentals · Security concepts · Cloud concepts

NoVA/DC Area Note: The Northern Virginia corridor is the largest data center market in the world — Ashburn hosts ~70% of global internet traffic. SOA-C02 and AZ-104 are the most in-demand admin certs in this market. Your Security+ (DoD 8570) already qualifies you for many government/contractor roles. An active security clearance, if you have one from your Navy service, is a major force multiplier here.